# Lesson 1: Factoring Review: Common Factors and $x^2 + bx + c$

*Taking out the greatest common factor, factoring by grouping, factoring $x^2 + bx + c$, and always factoring fully.*

## 1. The Greatest Common Factor

**Factoring** writes a sum or difference of terms as a **product**. It is the reverse of expanding. The first step in every factoring question is to look for the **greatest common factor (GCF)**: the largest number and the highest power of each variable that divide *every* term. Write the GCF in front and divide each term by it to get what goes in the brackets.

$\underbrace{6x^3 - 15x^2}_{\text{sum or difference}} = \underbrace{3x^2(2x - 5)}_{\text{product}}$ Check by expanding: $3x^2(2x) - 3x^2(5) = 6x^3 - 15x^2$ ✓

When the first term is negative, it is usually neater to take out a **negative** common factor. Every sign inside the brackets changes: $-4x^2 + 10x = -2x(2x - 5)$.

**Example 1**

Factor by taking out the GCF.

(a) $12x^3 - 18x^2$ (b) $10a^4b^2 + 15a^2b^3 - 5a^2b$ (c) $-4m^3 + 12m^2 - 8m$

## 2. A Binomial as the Common Factor

A common factor can be a whole bracket. In $4x(x - 3) + 7(x - 3)$ both terms contain $(x - 3)$, so take it out just like a monomial: $(x-3)(4x+7)$. Brackets that are **opposites**, like $(2-b)$ and $(b-2)$, are not the same yet: use $2-b = -(b-2)$ to make them match.

**Example 2**

Factor.

(a) $5y(y + 6) - 2(y + 6)$ (b) $3x(2x - 1) + (2x - 1)$ (c) $a(b - 2) + 7(2 - b)$

## 3. Factoring by Grouping

A polynomial with **four terms** often has no common factor in all four terms, but does in each **pair**. Group the terms in pairs, take the GCF out of each pair, and then take out the common bracket. If the brackets do not match, try taking a negative factor out of the second pair, or rearrange the terms.

$\underbrace{x^3 + 4x^2}_{x^2(x+4)}\ \underbrace{-3x - 12}_{-3(x+4)} = x^2(x + 4) - 3(x + 4) = (x + 4)(x^2 - 3)$

**Example 3**

Factor by grouping.

(a) $6ab - 9a + 4b - 6$ (b) $xy - 5x - 2y + 10$ (c) $2xy + 15 + 6x + 5y$

### 4. Factoring $x^2 + bx + c$

Expanding $(x+m)(x+n)$ gives $x^2 + (m+n)x + mn$. So to factor a trinomial with leading coefficient 1, find two integers $m$ and $n$ that **multiply to** $c$ and **add to** $b$.

$x^2 + bx + c = (x+m)(x+n)$ where $m \times n = c$ and $m + n = b$
Signs: $c > 0$ means $m$ and $n$ have the *same* sign (the sign of $b$); $c < 0$ means *opposite* signs (the larger size takes the sign of $b$).

List the factor pairs of $c$ in order until one pair has the right sum. Check every answer by expanding. The same method works with two variables: $x^2 + bxy + cy^2 = (x + my)(x + ny)$.

**Example 4**

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Factor. Check one answer by expanding.
(a) $n^2 + 13n + 36$ (b) $x^2 - 10x + 21$ (c) $y^2 + 2y - 35$ (d) $p^2 - 9p - 36$

**Example 5**

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Factor.
(a) $x^2 - 7xy + 12y^2$ (b) $a^2 + ab - 20b^2$

### 5. Factor Fully

To **factor fully**, keep going until no factor can be factored again. Always take out the GCF *first*: it makes the numbers smaller. If the $x^2$ term is negative, take out $-1$ (or a negative GCF) so the leading coefficient in the brackets is positive.

**Factoring checklist:** (1) GCF first (negative if the leading term is negative). (2) Count the terms: four terms → grouping; trinomial $x^2 + bx + c$ → product and sum. (3) Check each factor: can it be factored again? (4) Check by expanding.

A trinomial written in **ascending order** (constant first) can also be factored directly: $35 + 2x - x^2 = (7 - x)(5 + x)$. This equals $-(x - 7)(x + 5)$, since $7 - x = -(x - 7)$. Both forms are correct.

**Example 6**

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Factor fully.

(a) $4x^2 - 8x - 60$

(b) $2a^3 + 14a^2 + 24a$

(c) $-x^2 + 2x + 48$

(d) $5x^2y - 15xy^2 - 50y^3$

### 6. Trinomials That Do Not Factor, and Testing a Factor

If no pair of integers has product $c$ and sum $b$, the trinomial **does not factor over the integers**. Say so, and show the pairs you checked. To decide whether a binomial is a **factor** of a polynomial, factor the polynomial fully and see whether the binomial appears (a factor such as $-(x-7)$ or $(7-x)$ also counts as $x-7$).

**Example 7**

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(a) Show that $x^2 + 5x + 8$ does not factor over the integers.

(b) Find all integers $b$ so that $x^2 + bx + 10$ can be factored.

(c) Is $x - 4$ a factor of $2x^2 - 2x - 24$?

(d) Is $x + 2$ a factor of $x^3 + 2x^2 - 9x - 18$?

**Example 8**

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A rectangular patio has an area of $(x^2 + 11x + 28)$ m$^2$. Its length and width are binomials with integer coefficients.

(a) Write expressions for the length and the width.

(b) Write and simplify an expression for the perimeter.

(c) Find the dimensions, perimeter and area when $x = 5$. Check that length $\times$ width equals the area.

## Lesson 2: Factoring $ax^2 + bx + c$

*Factoring trinomials whose leading coefficient is not 1 by decomposition, including rational coefficients and perfect squares.*

### 1. The Method of Decomposition

When two binomials are expanded, the two middle products combine into the $x$-term. In $(2x + 3)(x + 4)$ the four parts of the area model are $2x^2$, $8x$, $3x$ and 12, so the trinomial is $2x^2 + 11x + 12$.

| | $x$ | 4 |
|---|---|---|
| $2x$ | $2x^2$ | $8x$ |
| 3 | $3x$ | 12 |

The two middle parts multiply to $8x \cdot 3x = 24x^2$, and $(2x^2)(12) = 24x^2$ too. So the middle numbers 8 and 3 multiply to $ac = 2 \times 12 = 24$ and add to $b = 11$.

This is always true, so to factor $ax^2 + bx + c$ we **decompose** (split) the middle term using two numbers with product $ac$ and sum $b$, then factor by grouping.

**Decomposition** for $ax^2 + bx + c$: (1) Find $m$ and $n$ with $m \times n = ac$ and $m + n = b$. (2) Rewrite $bx$ as $mx + nx$. (3) Factor the four terms by grouping. (4) Check by expanding.

**Example 1**

(a) Expand $(3x + 2)(2x + 5)$, showing all four products before you collect like terms.

(b) For $6x^2 + 19x + 10$: $ac =$ _____. The two middle coefficients from (a) are _____ and _____: their product is _____ and their sum is _____.

**Example 2**

Factor by decomposition.

(a) $3x^2 + 10x + 8$ (b) $5x^2 + 17x + 6$ (c) $4x^2 + 16x + 15$

### 2. Negative Terms, and Guess and Check

The sign rules from Lesson 1 still apply, now to $ac$: if $ac > 0$ the two numbers have the same sign as $b$; if $ac < 0$ they have opposite signs. When you group, take a negative factor out of the second pair if needed so the brackets match. If no pair of integers works, the trinomial **does not factor over the integers**.

**Example 3**

Factor by decomposition.

(a) $4x^2 - 4x - 15$ (b) $6a^2 - 13a + 6$ (c) $5y^2 + 13y - 6$

**Guess and check** is another method. Choose factors of $a$ for the first terms of the binomials and factors of $c$ for the last terms, expand mentally, and adjust until the middle term is right. It is fast when $a$ and $c$ have few factors; decomposition is more reliable when they have many.

**Example 4**

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Factor $3x^2 - 7x - 6$ by guess and check.

### 3. GCF First, and Negative Leading Coefficients

Always take out the GCF first: the trinomial left is smaller and easier to decompose, and the answer is then fully factored. If $a$ is negative, take out $-1$ (or the negative GCF) first. A trinomial in ascending order can be factored either way, as in Lesson 1.

**Example 5**

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Factor fully.

(a) $12x^2 + 2x - 4$ (b) $-6x^2 + 7x + 3$

(c) $8 + 2x - 3x^2$ (d) $18x^3 - 33x^2 - 30x$

### 4. Two Variables and Rational Coefficients

For $ax^2 + bxy + cy^2$, decompose exactly as before; each binomial has an $x$-term and a $y$-term.

**Example 6**

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Factor.

(a) $3x^2 + 7xy - 6y^2$ (b) $10a^2 - 23ab + 12b^2$

The coefficients $a$, $b$ and $c$ may be **fractions**. Take out a fractional common factor so that integers are left in the brackets: dividing each term by $\frac{1}{2}$ is the same as multiplying it by 2. When the denominators differ, take out $\frac{1}{\text{LCD}}$.

$\frac{1}{2}x^2 - x - 4 = \frac{1}{2}\left(x^2 - 2x - 8\right) = \frac{1}{2}(x-4)(x+2)$ Check: $\frac{1}{2} \times x^2$, $\frac{1}{2} \times (-2x)$, $\frac{1}{2} \times (-8)$ give back the original terms.

**Example 7**

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Factor fully.

(a) $\frac{1}{2}x^2 + x - 12$ (b) $\frac{3}{4}x^2 - 3x - 24$ (c) $\frac{1}{2}x^2 + \frac{5}{4}x + \frac{1}{2}$

**5. Perfect-Square Trinomials**

Squaring a binomial gives a **perfect-square trinomial**:

$A^2 + 2AB + B^2 = (A+B)^2$ $A^2 - 2AB + B^2 = (A-B)^2$

Test: the first and last terms are perfect squares ($A^2$ and $B^2$, both positive), and the middle term is $\pm 2AB$.

Decomposition also works on these, but recognizing the pattern is faster. A trinomial like $x^2 + 6x - 9$ is *not* a perfect square: its last term is negative.

**Example 8**

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(a) Which are perfect-square trinomials? Write those in the form $(A \pm B)^2$.

(i) $9x^2 + 12x + 4$ (ii) $x^2 - 10x + 16$

(iii) $25x^2 - 20xy + 4y^2$ (iv) $4x^2 + 6x + 9$

(b) Fill in the blank so that each trinomial is a perfect square.

$x^2 - _____ + 81$ $16x^2 + 40x + _____$ $_____ + 42x + 49$

**Example 9**

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(a) Factor $64x^2 - 48x + 9$.

(b) Factor fully: $2x^3 + 20x^2 + 50x$.

(c) A square floor tile has an area of $(4x^2 + 28x + 49)$ cm$^2$. Find an expression for its side length, then its side and perimeter when $x = 3$.

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## Lesson 3: Factoring by Substitution: $a(f(x))^2 + b(f(x)) + c$

*Spotting a quadratic pattern inside a bigger expression, replacing the repeated part with one letter, factoring, then substituting back and simplifying.*

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### 1. Trinomials in Quadratic Form

In $x^2 + bx + c$ the powers of $x$ are 2, 1 and 0. Many other trinomials have the same shape: the variable part of the first term is the **square** of the variable part of the middle term, and the last term is a constant. Such an expression is in **quadratic form**:

$$x^4 - 7x^2 + 12 = (x^2)^2 - 7(x^2) + 12 \qquad x^6 + 2x^3 - 8 = (x^3)^2 + 2(x^3) - 8$$

Replace the repeated expression with a single letter, factor the simpler trinomial, then put the expression back. This is **factoring by substitution**.

**Factoring $a(f(x))^2 + b(f(x)) + c$ by substitution**
1. Find the repeated expression $f(x)$ and write the trinomial as $a(\ \cdot\ )^2 + b(\ \cdot\ ) + c$. 2. Let $A = f(x)$. 3. Factor $aA^2 + bA + c$.
4. Replace $A$ with $f(x)$. 5. Simplify inside each bracket and factor each factor fully.

**Example 1**

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Factor.
(a) $x^4 + 9x^2 + 14$ (b) $y^4 - 3y^2 - 28$ (c) $x^6 + 4x^3 - 21$

With practice the letter $A$ can be skipped: think "two numbers with product 14 and sum 9" and write the $x^2$ factors directly.

### 2. Finishing with a Difference of Squares

After substituting back, a factor such as $x^2 - 9$ can often be factored again. Recall the **difference of squares** pattern from Math 10C (Lesson 4 develops it fully):

$A^2 - B^2 = (A - B)(A + B)$ e.g. $x^2 - 9 = (x - 3)(x + 3)$, $4x^2 - 1 = (2x - 1)(2x + 1)$.
A **sum** of squares such as $x^2 + 9$ does not factor. Neither does $x^2 - 7$ over the integers (7 is not a perfect square).

To **factor fully**: take out the GCF first, and check every factor at the end.

**Example 2**

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Factor fully.

(a) $x^4 - 13x^2 + 36$ (b) $m^4 - 5m^2 - 36$ (c) $3y^5 - 54y^3 + 243y$

### 3. Leading Coefficient Not 1: Decomposition

When $a \neq 1$, factor $aA^2 + bA + c$ by **decomposition** (Lesson 2): find two numbers with product $ac$ and sum $b$, split the middle term, and group. You can also split the middle term without writing $A$ at all.

**Example 3**

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Factor $3x^4 - 10x^2 - 8$ fully two ways.

(a) Substitute $A = x^2$ first. (b) Split $-10x^2$ directly.

**Example 4**

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Factor fully.

(a) $9x^4 - 37x^2 + 4$ (b) $6a^2b^2 - 5ab - 6$ (c) $2p^6 + 5p^3 - 12$

### 4. Substituting for a Binomial

The repeated expression can be a binomial, as in $(x + 2)^2 - 3(x + 2) - 10$. Let $A = x + 2$, factor, then replace $A$ with $(x + 2)$ **in brackets**, simplify each factor and take out any common factor.

**Example 5**

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Factor fully.

(a) $(x+2)^2 - 3(x+2) - 10$ (b) $2(3x-1)^2 + 5(3x-1) - 3$

You could instead **expand first** and factor the result. It gives the same answer but takes longer, and the expanded trinomial usually has bigger numbers.

**Example 6**

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Factor $4(x-1)^2 - 4(x-1) - 3$ two ways.

(a) By substitution. (b) By expanding first.

**Example 7**

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A trinomial in quadratic form can be a **perfect square**. Factor fully.

(a) $(2x+1)^2 - 6(2x+1) + 9$ (b) $x^4 + 10x^2 + 25$

### 5. Other Repeated Expressions

Any expression can play the role of $A$. In trigonometry, $\sin^2 x$ means $(\sin x)^2$, so $2\sin^2 x + \sin x - 1$ is a trinomial in $\sin x$. An expression can also contain **two** repeated binomials, like $aA^2 + bAB + cB^2$: factor it as you would $6x^2 + xy - 2y^2$, then substitute both back.

**Example 8**

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Factor fully.

(a) $2\sin^2 x + \sin x - 1$

(b) $2(x+2)^2 - 3(x+2)(2x-1) - 2(2x-1)^2$

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## Lesson 4: Differences of Squares, Including $a^2(f(x))^2 - b^2(g(y))^2$

*Recognizing one perfect square subtracted from another, factoring it into conjugate binomials, and using the pattern with fractions, higher powers and binomials.*

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### 1. The Difference of Squares Pattern

Multiply two binomials that differ only in the sign between their terms:

$$(A-B)(A+B) = A^2 + AB - AB - B^2 = A^2 - B^2$$

The middle terms cancel. Read backwards, this is a factoring pattern.

**Difference of squares:** $A^2 - B^2 = (A-B)(A+B)$
It applies when there are **two terms**, both are **perfect squares**, and they are **subtracted**. Find $A$ and $B$ by taking the square root of each term: $49x^2 = (7x)^2$, $25y^2 = (5y)^2$, so $49x^2 - 25y^2 = (7x-5y)(7x+5y)$.

A **sum of squares** $A^2 + B^2$ does **not** factor over the real numbers. For $x^2 + 36$ we would need two numbers with sum 0 and product +36; numbers with sum 0 are opposites, and opposites have a negative product.

**Example 1**

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Factor, if possible.

(a) $25x^2 - 64$

(b) $49a^2 - 81b^2$

(c) $100m^2n^2 - 9p^2$

(d) $x^2 + 36$

(e) $-y^2 + 4x^2$

(f) $1 - \sin^2 x$

### 2. Common Factor First

Always look for a GCF first. Numbers like 12 and 27 are not perfect squares, but after removing the common factor 3 the binomial is $4x^2 - 9y^2$. If the first term is negative, taking out a negative common factor makes the pattern easier to see.

**Example 2**

Factor fully.

(a) $12x^2 - 27y^2$

(b) $5a^3b - 45ab^3$

(c) $8x^2 + 50$

(d) $-18x^2 + 8$

**Example 3**

A rectangular banner has an area of $36x^2 - 25y^2$ square centimetres. Its length and width are both of the form $ax + by$, where $a$ and $b$ are integers.

(a) Write expressions for the length and the width.

(b) The perimeter is 96 cm. Determine $x$.

(c) The length is also 3 times the width. Determine $y$ and the dimensions of the banner.

**3. Rational Coefficients**

A fraction is a perfect square when its numerator and denominator are both perfect squares: $\frac{1}{4}x^2 = \left(\frac{1}{2}x\right)^2$ and $\frac{y^2}{25} = \left(\frac{y}{5}\right)^2$. Decimals work too: $0.09 = 0.3^2$. Another way is to take out a fractional common factor first, as in Lesson 2.

**Example 4**

Factor fully.

(a) $\frac{1}{4}x^2 - 9$ (two ways)

(b) $\frac{x^2}{16} - \frac{y^2}{25}$

(c) $0.09a^2 - 0.49$

### 4. Higher Powers and Repeated Factoring

Even powers are perfect squares: $x^4 = (x^2)^2$, $x^6 = (x^3)^2$, $16y^4 = (4y^2)^2$. After one use of the pattern, the *difference* factor may be a difference of squares again, so keep going. The *sum* factor stays.

**Example 5**

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Factor fully.

(a) $x^4 - 81$

(b) $16a^4 - b^4$

(c) $3x^4 - 48y^4$

(d) $m^8 - 1$

(e) $x^6 - 49$

### 5. Squares of Binomials: $a^2(f(x))^2 - b^2(g(y))^2$

The squared quantities can be binomials. Let $A = a\,f(x)$ and $B = b\,g(y)$, use $(A - B)(A + B)$, then **keep the brackets** while you substitute back (the minus sign applies to every term of $B$), simplify each factor and take out any common factor.

$$4(2x-1)^2 - 9(y+2)^2 = \big[2(2x-1)\big]^2 - \big[3(y+2)\big]^2 \qquad A = 2(2x-1),\ B = 3(y+2)$$

**Example 6**

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Factor fully.

(a) $(x+3)^2 - 25$

(b) $16 - (y-1)^2$

(c) $4(2x-1)^2 - 9(y+2)^2$

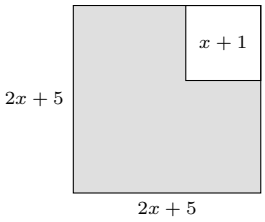
(d) $(3x+1)^2 - (x-5)^2$

**Example 7**

A square patio has side length $(2x + 5)$ m. A square planter with side length $(x + 1)$ m is cut out of it.

(a) Write an expression for the area of the patio that remains.

(b) Factor it fully using a difference of squares.




(c) Check by expanding each square first, then factoring.

(d) Find the remaining area when $x = 3$.

### 6. Choosing a Method

**Factoring checklist**
1. Take out the **GCF** (a negative one if the first term is negative).
2. Count the terms. **Two:** a difference of squares? (A sum of squares does not factor.)
   **Three:** $x^2 + bx + c$; $ax^2 + bx + c$ by decomposition; a perfect square; quadratic form (substitution).
   **Four:** grouping (two and two, or three and one to make a difference of squares).
3. Check every factor and factor again if you can. 4. Check by expanding.

A binomial is a **factor** of a polynomial if it appears in the polynomial's fully factored form (or is a product of factors that appear there).

**Example 8**

Factor fully.

(a) $2x^3 - 18x$

(b) $x^3 + 2x^2 - 9x - 18$

(c) $4x^4 - 37x^2 + 9$

(d) $6x^2 - 7x - 20$

(e) $x^2 - 6x + 9 - y^2$

(f) $3a^2 + 12$

**Example 9**

Determine whether each binomial is a factor of $3x^4 - 48$. (a) $x - 2$ (b) $x - 4$ (c) $x^2 + 4$

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## Lesson 5: Solving Quadratic Equations by Factoring

*Using the zero product property to find roots, writing an equation from its roots, and solving problems that lead to a quadratic equation.*

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### 1. The Zero Product Property

If a product of numbers is zero, at least one of the numbers must be zero. This is the **zero product property**.

| **Zero product property:** if $ab = 0$, then $a = 0$ or $b = 0$. |
|---|

So when one side of an equation is a product of factors and the other side is 0, set each factor equal to 0 and solve. The values that make the equation true are its **roots** (or solutions). A constant factor such as 5 or $-2$ can never be 0, so it gives no root.

**Example 1**

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Solve each equation.

(a) $(x + 6)(x - 4) = 0$

(b) $4x(x - 9) = 0$

(c) $(2x - 7)(3x + 4) = 0$

(d) $-5(x + 1)(x - 8) = 0$

### 2. Solving a Quadratic Equation in Standard Form

A **quadratic equation** can be written in **standard form** $ax^2 + bx + c = 0$, $a \neq 0$. The zero product property only works when one side is 0, so:

| **To solve by factoring:** (1) move every term to one side so the other side is 0; (2) divide out (or factor out) any common constant factor, and clear fractions by multiplying by the LCD; (3) factor fully; (4) set each factor with a variable equal to 0 and solve; (5) check in the original equation. |
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Never divide both sides by $x$ (or by any factor with a variable): that factor could be 0, and you lose a root. Factor it out instead.

**Example 2**

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Solve.

(a) $x^2 + 3x - 28 = 0$

(b) $16x^2 - 25 = 0$

(c) $6x^2 = 21x$

(d) $3x^2 - 48 = 0$

**Example 3**

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Solve. A root that comes from a squared factor is a **double root** (the equation has only one distinct root).

(a) $4x^2 - 4x - 15 = 0$

(b) $-2x^2 + 10x = 12$

(c) $9x^2 + 4 = 12x$

(d) $\frac{1}{4}x^2 - x - 3 = 0$

**Example 4**

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Expand and simplify first, then solve.

(a) $(x + 2)(x - 5) = 8$

(b) $2x(x + 3) = (x + 1)^2 + 11$

(c) $(x - 4)(x + 1) = 2(x - 4)$

### 3. Roots, Zeros and $x$-Intercepts

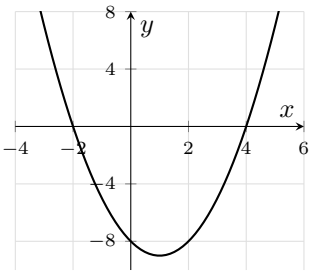
The **roots** of the equation $ax^2 + bx + c = 0$, the **zeros** of the function $y = ax^2 + bx + c$ (the $x$-values that make $y = 0$) and the **$x$-intercepts** of its graph (a parabola) are the same numbers. A double root means the parabola just touches the $x$-axis. (Unit 5 develops these graphs.)

**Example 5**

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The graph of $y = x^2 - 2x - 8$ is shown.
(a) Read the $x$-intercepts from the graph.

(b) Confirm by solving $x^2 - 2x - 8 = 0$ algebraically, and state the zeros of the function.




### 4. Writing an Equation from Its Roots

Work backwards: a root $r$ comes from the factor $(x - r)$. A fractional root $x = \frac{p}{q}$ comes from the factor $(qx - p)$, which keeps the coefficients integers.

If the roots are $r_1$ and $r_2$, an equation is $(x - r_1)(x - r_2) = 0$. Root $x = \frac{2}{3} \Rightarrow$ factor $(3x - 2)$; root $x = -\frac{1}{4} \Rightarrow$ factor $(4x + 1)$.

Multiplying the equation by any non-zero constant gives an equation with the same roots, so there are many correct answers; give the one with the smallest integer coefficients.

**Example 6**

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Write a quadratic equation in standard form, with integer coefficients, that has the given roots.

(a) 5 and $-3$

(b) $-\frac{1}{2}$ and 4

(c) 0 and $-\frac{7}{3}$

(d) a double root of $\frac{3}{5}$

### 5. Problems That Lead to a Quadratic Equation

1. Let a variable stand for an unknown (draw a diagram when there is a shape) and write the other unknowns in terms of it.
2. Write an equation from the information, put it in standard form and solve it by factoring.
3. **Reject inadmissible roots**: a length, area or time cannot be negative, a count must be a whole number, and a dimension cannot make another dimension negative.
4. Answer the question asked, with units.

**Example 7**

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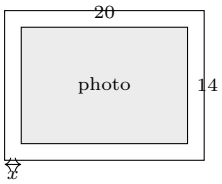
(a) The product of two consecutive odd integers is 143. Determine the integers.

(b) Twice the square of a number is 15 more than the number. Determine the number.

**Example 8**

(a) A rectangular garden is 3 m longer than twice its width. Its area is 65 $m^2$. Determine its dimensions and its perimeter.

(b) A 20 cm by 14 cm photo has a frame of uniform width $x$ cm around it. The photo and frame together cover 432 $cm^2$. Determine $x$.




**Example 9**

A ball is thrown upward from a balcony. Its height above the ground after $t$ seconds is $h = -5t^2 + 20t + 25$ metres.

(a) When does the ball hit the ground?

(b) When is the ball 40 m above the ground?

## Lesson 6: Solving Radical Equations and Checking for Extraneous Roots

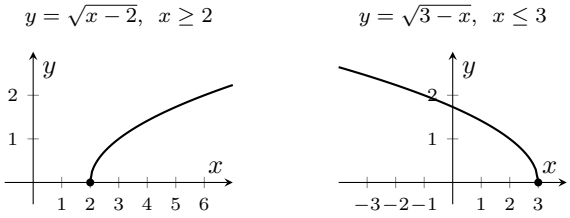
*Stating restrictions, isolating and squaring a square root, and verifying every answer to catch extraneous roots.*

### 1. Restrictions on the Variable

A **radical equation** has the variable under a radical sign. In this course the radicals are square roots. A square root of a negative number is not a real number, so the **radicand** must be non-negative:

> $\sqrt{A}$ is defined only when $A \geq 0$. To find the **restriction**, solve the inequality "radicand $\geq 0$". Dividing or multiplying an inequality by a negative number reverses the sign.

The restriction is the domain of the graph: $y = \sqrt{x-2}$ starts at $x = 2$ and goes right; $y = \sqrt{3-x}$ starts at $x = 3$ and goes left. When an equation has two radicals, $x$ must satisfy both restrictions at once.



**Example 1**

State the restriction on the variable.

(a) $\sqrt{x+4} = 6$

(b) $\sqrt{2x-5} + 1 = 4$

(c) $\sqrt{6-3x} = 9$

(d) $\sqrt{x-1} + \sqrt{5-x} = 2$

(e) $\sqrt{x} = \sqrt{x+3} - 1$

### 2. Solving by Squaring and Verifying

Squaring undoes a square root: $\left(\sqrt{A}\right)^2 = A$ when $A \geq 0$. Square both *sides*, not each term: get the radical alone first. If there is a number in front of the radical, divide by it first (or square it too: $\left(2\sqrt{A}\right)^2 = 4A$).

**Solving a radical equation:** (1) state the restriction; (2) isolate the radical; (3) square both sides; (4) solve; (5) **verify** each answer by substituting it into the *original* equation (left side = right side).

**Example 2**

Solve and verify.

(a) $\sqrt{3x+1} = 5$

(b) $2\sqrt{x-1} + 3 = 9$

(c) $\sqrt{4x-3} - 7 = -2$

(d) $\sqrt{\dfrac{x+4}{3}} = 2$

### 3. Extraneous Roots

The symbol $\sqrt{\ }$ means the **principal** (non-negative) square root: $\sqrt{9} = 3$, never $-3$. So an equation like $\sqrt{x} = -3$ can have no solution.

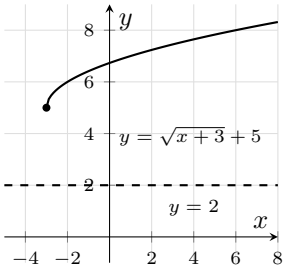
Squaring is not a reversible step. If $a = b$ then $a^2 = b^2$, but $a^2 = b^2$ does not mean $a = b$: $3^2 = (-3)^2$ and $3 \neq -3$. The squared equation can therefore have a root that the original equation does not have. A value found by correct algebra that does **not** satisfy the original equation is an **extraneous root**. It must be rejected, which is why every answer is verified.

After isolating: if $\sqrt{A}$ equals a **negative number**, there is no solution (squaring would produce an extraneous root). A value that breaks a **restriction** is also extraneous.

**Example 3**

(a) Solve $\sqrt{x} = -3$ by squaring. Verify, and state the solution.

(b) Solve $\sqrt{x+3} + 5 = 2$ and verify.




(c) The graphs of $y = \sqrt{x+3} + 5$ and $y = 2$ are shown. How do they confirm your answer to (b)?

### 4. A Radical on Each Side

When each side is a single radical, square both sides at once: $\left(\sqrt{A}\right)^2 = \left(\sqrt{B}\right)^2$ gives $A = B$. A coefficient in front of a radical is squared too. Both radicands must be non-negative, so check the answer in *both* radicands: a value that makes either one negative is extraneous.

**Example 4**

Solve and verify.

(a) $\sqrt{5x-2} = \sqrt{2x+7}$ (b) $3\sqrt{x-2} = \sqrt{5x+6}$ (c) $\sqrt{2x-9} = \sqrt{x-7}$

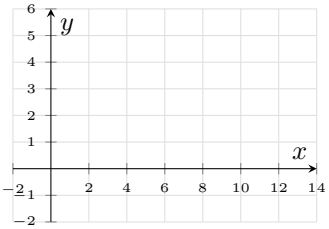
### 5. Checking with a Graph

A graphing calculator can solve or check a radical equation. Enter the left side as $Y_1$ and the right side as $Y_2$, press GRAPH, then use 2nd [TRACE] (CALC), 5: intersect. The $x$-coordinate of each point of intersection is a root. If the graphs do not meet, there is no solution; an extraneous root shows up as an $x$-value where the graphs do *not* meet. Graphing also gives decimal approximations when a root is not a whole number.

**Example 5**

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Consider the equation $\sqrt{2x-4} = 4$.
(a) Solve algebraically and verify.




(b) Graph $Y_1 = \sqrt{2x-4}$ and $Y_2 = 4$ in the window $x$: $[-2, 14, 2]$, $y$: $[-2, 6, 1]$. Sketch the display and mark the point of intersection.

### 6. Problems Modelled by a Radical Equation

Many formulas contain a square root. To find the variable under the radical, substitute the known values, isolate the radical, square both sides and solve. Round only at the end, as the question asks.

**Example 6**

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A police officer estimates a car's speed from the length of its skid marks with the formula $s = 4\sqrt{10d}$, where $s$ is the speed in km/h and $d$ is the length of the skid marks in metres. How long are the skid marks of a car travelling at 80 km/h?

**Example 7**

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The period of a pendulum (the time for one full swing, in seconds) is $T = 2\pi\sqrt{\dfrac{L}{9.8}}$, where $L$ is its length in metres.

(a) Determine the period of a 1.5 m pendulum, to the nearest hundredth of a second.

(b) Determine the length that gives a period of 2.5 s, to the nearest centimetre.

(c) Solve the formula for $L$.

## Lesson 7: Radical Equations That Lead to Quadratics

*Squaring a binomial side correctly, solving the quadratic by factoring, and checking every root — including equations that must be squared twice.*

### 1. Before You Square: Binomial Squares and Restrictions

When a radical equation has a variable outside the radical, squaring both sides squares a **binomial**. Use the pattern for the square of a binomial; never square each term separately.

$(a+b)^2 = a^2 + 2ab + b^2$ $\quad (a-b)^2 = a^2 - 2ab + b^2$ $\quad (x-2)^2 = x^2 - 4x + 4$, **not** $x^2 + 4$.

A radical squared gives its radicand: $\left(\sqrt{A}\right)^2 = A$, and $\left(k\sqrt{A}\right)^2 = k^2 A$.

**Example 1**

Expand and simplify. Assume each radicand is non-negative.

(a) $(x-3)^2$

(b) $(2x+5)^2$

(c) $(6-x)^2$

(d) $(2+\sqrt{x})^2$

(e) $(3\sqrt{x})^2$

(f) $\left(\sqrt{x+1}-3\right)^2$

**Restrictions.** A square root is never negative, so in $\sqrt{A} = B$ two things must be true: the radicand $A \geq 0$, and the other side $B \geq 0$. Isolate the radical first, then write both conditions and combine them.

**Example 2**

State the restrictions on $x$ for each equation.

(a) $\sqrt{x+12} = x$

(b) $\sqrt{3x+4} = x-2$

(c) $\sqrt{10-3x} + 2 = 4 - x$

### 2. Solving: Isolate, Square, Factor, Check

**Solving a radical equation.** (1) State the restrictions. (2) Isolate the radical. (3) Square both sides. (4) Write the quadratic in standard form ($ax^2 + bx + c = 0$), factor, and use the zero product property. (5) **Check every root in the original equation.** A root that does not satisfy the original equation is an **extraneous root**: reject it.

Squaring can create extraneous roots because two *different* numbers can have the same square: $3^2 = (-3)^2$, but $3 \neq -3$. The restrictions predict which roots will fail, but the check by substitution is still required.

**Example 3**

Solve $\sqrt{x+12} = x$ and verify.

**Example 4**

Solve and verify.

(a) $\sqrt{2x+7} - x = 2$

(b) $2\sqrt{x+1} = x - 2$

(c) $3 + \sqrt{2x-3} = 2x$

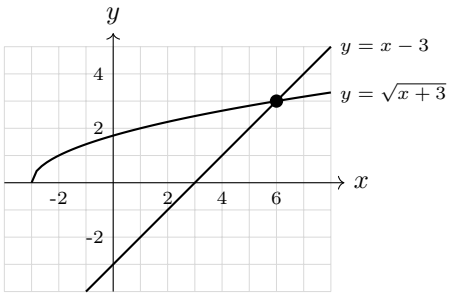
### 3. Why Extraneous Roots Appear: A Graph

The roots of $\sqrt{A} = B$ are the $x$-coordinates of the points where the graphs of $y = \sqrt{A}$ and $y = B$ meet. After squaring, $A = B^2$ is *also* true when $\sqrt{A} = -B$, so the squared equation picks up the intersections with $y = -B$ as well. Those are the extraneous roots. To check on a calculator, graph $Y_1$ = left side and $Y_2$ = right side and use the intersect feature.

**Example 5**

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(a) Solve $\sqrt{x+3} = x - 3$ algebraically.




(b) The graphs of $y = \sqrt{x+3}$ and $y = x-3$ are shown. Explain how the graph shows that one root is extraneous.

### 4. Equations Like $x + \sqrt{x} = 12$

When the variable appears both inside and outside the radical, isolate the radical and square as usual. Another way is **substitution**: let $u = \sqrt{x}$ (so $u \geq 0$ and $x = u^2$), factor the quadratic in $u$, reject any negative $u$, then square to get $x$.

**Example 6**

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Solve $x + \sqrt{x} = 12$ in two ways.

(a) Isolate the radical and square.

(b) Substitute $u = \sqrt{x}$.

### 5. Two Radicals: Squaring Twice

If there are two radicals, isolate one of them (usually the more complicated one) and square. One radical will remain: isolate it and square again. Then solve and check. When you square $1 + \sqrt{x}$, the middle term $2\sqrt{x}$ is the radical that remains.

**Example 7**

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Solve and verify.

(a) $\sqrt{x+7} - \sqrt{x} = 1$

(b) $\sqrt{x-1} - \sqrt{2x-4} = 1$

### 6. Problems Modelled by Radical Equations

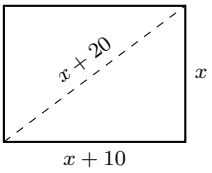
Let a variable stand for the unknown, write the radical equation, solve and check it, then answer in a sentence. A root can satisfy the equation and still be **inadmissible** in the context (a negative length, for example): reject it for that reason, not because it is extraneous.

**Example 8**

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(a) A positive number decreased by its square root is 30. Determine the number.

(b) A rectangular cutting board is $x$ cm wide. Its length is 10 cm more than its width, and its diagonal is 20 cm more than its width. Write a radical equation for the diagonal, then determine the dimensions of the board.




## Lesson 8: Unit 4 Review

*Every factoring method, solving by factoring and solving radical equations on one page, mixed examples that combine them, and a review assignment written like the unit test.*

### 1. Unit Summary

| Method | When to use it | Example |
|---|---|---|
| Greatest common factor (GCF) | Always first. Take out a negative GCF if the leading term is negative. A binomial can be the common factor. | $-6x^2 + 18x = -6x(x-3)$<br>$5a(a-2) + 3(a-2) = (a-2)(5a+3)$ |
| Grouping | Four terms: group in pairs, factor each pair, take out the common binomial. | $x^3 - 4x^2 + 3x - 12 = x^2(x-4) + 3(x-4)$<br>$= (x-4)(x^2+3)$ |
| $x^2 + bx + c$ | Two numbers with product $c$ and sum $b$. Works with two variables. | $x^2 - 3xy - 28y^2 = (x-7y)(x+4y)$ |
| $ax^2 + bx + c$ (decomposition) | Two numbers with product $ac$ and sum $b$; split the middle term and group. Factor out a fraction for rational coefficients. | $6x^2 + 7x - 3$: $ac = -18$, $9 + (-2) = 7$<br>$= 3x(2x+3) - (2x+3) = (2x+3)(3x-1)$ |
| Perfect square trinomial | First and last terms are squares and the middle term is $2AB$. | $9x^2 - 30x + 25 = (3x-5)^2$ |
| Difference of squares | $A^2 - B^2 = (A-B)(A+B)$. A sum of squares does not factor. | $x^4 - 81 = (x^2-9)(x^2+9)$<br>$= (x-3)(x+3)(x^2+9)$ |
| Substitution | A repeated expression $f(x)$ or $x^2$: let $A = f(x)$, factor, substitute back, simplify each factor. | $(x+1)^2 - 5(x+1) + 6 = (A-2)(A-3)$<br>$= (x-1)(x-2)$ |

**Factoring checklist.** (1) GCF first. (2) Count the terms: two → difference of squares; three → $x^2 + bx + c$, decomposition, perfect square, or substitution; four → grouping. (3) Check every factor and factor again if you can (**factor fully**). (4) Check by expanding. A binomial is a factor of a polynomial if it appears in the fully factored form, or is a product of factors that do.

**Solving by factoring.** Move every term to one side ($ax^2 + bx + c = 0$), divide out a common factor, factor, and use the **zero product property**: if $AB = 0$ then $A = 0$ or $B = 0$. Never divide both sides by an expression containing the variable (a root is lost). The **roots** of $ax^2 + bx + c = 0$, the **zeros** of $y = ax^2 + bx + c$ and the $x$**-intercepts** of its graph are the same numbers. Roots $r_1$ and $r_2$ give the equation $(x - r_1)(x - r_2) = 0$; a root $\frac{p}{q}$ gives the factor $(qx - p)$. In a word problem, reject roots that are **inadmissible** (a negative length or time).

**Radical equations.** (1) State the **restrictions**: each radicand $\geq 0$, and the side opposite an isolated radical $\geq 0$. (2) Isolate a radical. (3) Square both sides (square a binomial side as $(a-b)^2 = a^2 - 2ab + b^2$). (4) If a radical remains, isolate it and square again. (5) Solve (often a quadratic by factoring). (6) **Verify** every root in the original equation. Squaring can create an **extraneous root** because $a^2 = b^2$ does not mean $a = b$ (for example $3^2 = (-3)^2$).

**Common errors.** $(x-2)^2 \neq x^2 + 4$. $(2x-4)(x+2)$ is not fully factored: $2(x-2)(x+2)$. $x^2 + 9$ does not factor. Dividing $x^2 = 5x$ by $x$ loses the root $x = 0$. A root of the squared equation is not automatically a root of the radical equation.

## 2. Factoring

### Example 1

Factor fully.

(a) $4x^3 - 8x^2 - 60x$

(b) $10x^2 - 11xy - 6y^2$

(c) $x^3 + 2x^2 - 9x - 18$

(d) $\frac{1}{3}x^2 - 2x - 9$

### Example 2

Factor fully.

(a) $3(x-1)^2 + 4(x-1) - 4$

(b) $9x^4 - 37x^2 + 4$

(c) $16x^4 - 81y^4$

(d) $25(x+1)^2 - 4(x-2)^2$

(e) $\dfrac{x^2}{9} - \dfrac{y^2}{4}$

**Example 3**

(a) Is $x + 4$ a factor of $2x^2 + 5x - 12$?

(b) Is $3x + 2$ a factor of $6x^3 + x^2 - 2x$?

(c) Find every integer $k$ so that $x^2 + kx - 12$ can be factored over the integers.

(d) Find $k$ so that $9x^2 + kx + 49$ is a perfect square trinomial.

### 3. Solving Quadratic Equations

**Example 4**

Solve by factoring.

(a) $6x^2 = 7x + 3$

(b) $(x - 4)(x + 1) = 6$

(c) $3x^2 - 48 = 0$

(d) $4x^2 + 20x + 25 = 0$

**Example 5**

(a) Determine the zeros of $y = 2x^2 - x - 6$ and the $x$-intercepts of its graph.

(b) Write a quadratic equation with integer coefficients whose roots are $\frac{2}{3}$ and $-5$.

**Example 6**

A 20 cm by 30 cm photo is surrounded by a frame of uniform width $x$ cm. The photo and frame together cover 1064 cm$^2$.

Determine the width of the frame.

## 4. Radical Equations

### Example 7

Solve and verify.

(a) $3\sqrt{2x-1}-2=7$

(b) $\sqrt{x+8}+4=x$

### Example 8

Consider the equation $\sqrt{2x+1}-\sqrt{x+4}=1$.

(a) State the restrictions on $x$.

(b) Explain why an extraneous root may appear when this equation is solved algebraically.

(c) Solve the equation and verify.